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Design and Analysis of Algorithms (E1ITB303)

Micro Project Report

On

Maximizing Train Throughput Using Graph Algorithms and Greedy Scheduling

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Abstract

Introduction

1.1 Project Overview

This project models a railway network as a graph and applies algorithmic techniques to optimize train movement. The goal is to improve efficiency by reducing delays and avoiding conflicts.

1.2 Objectives of the Project

- To model a railway system using graph representation
- To find shortest paths using Dijkstra's algorithm
- To handle conflicts using greedy scheduling
- To reduce waiting time and improve throughput

1.3 Problem Statement

In a railway network, multiple trains share limited track sections, which can lead to conflicts and delays. The challenge is to design an algorithmic system that efficiently schedules trains, avoids collisions, and minimizes total waiting time.

1.4 Scope of the Project

This project focuses on:

- Graph-based modeling of railway systems
- Shortest path computation
- Conflict resolution using scheduling logic
- It does not include real-time signaling or large-scale railway simulation.

Algorithm Design

2.1 Description of the Algorithm

The system combines two main approaches:

1. Dijkstra's Algorithm

Used to find the shortest path between source and destination.

2. Greedy Scheduling

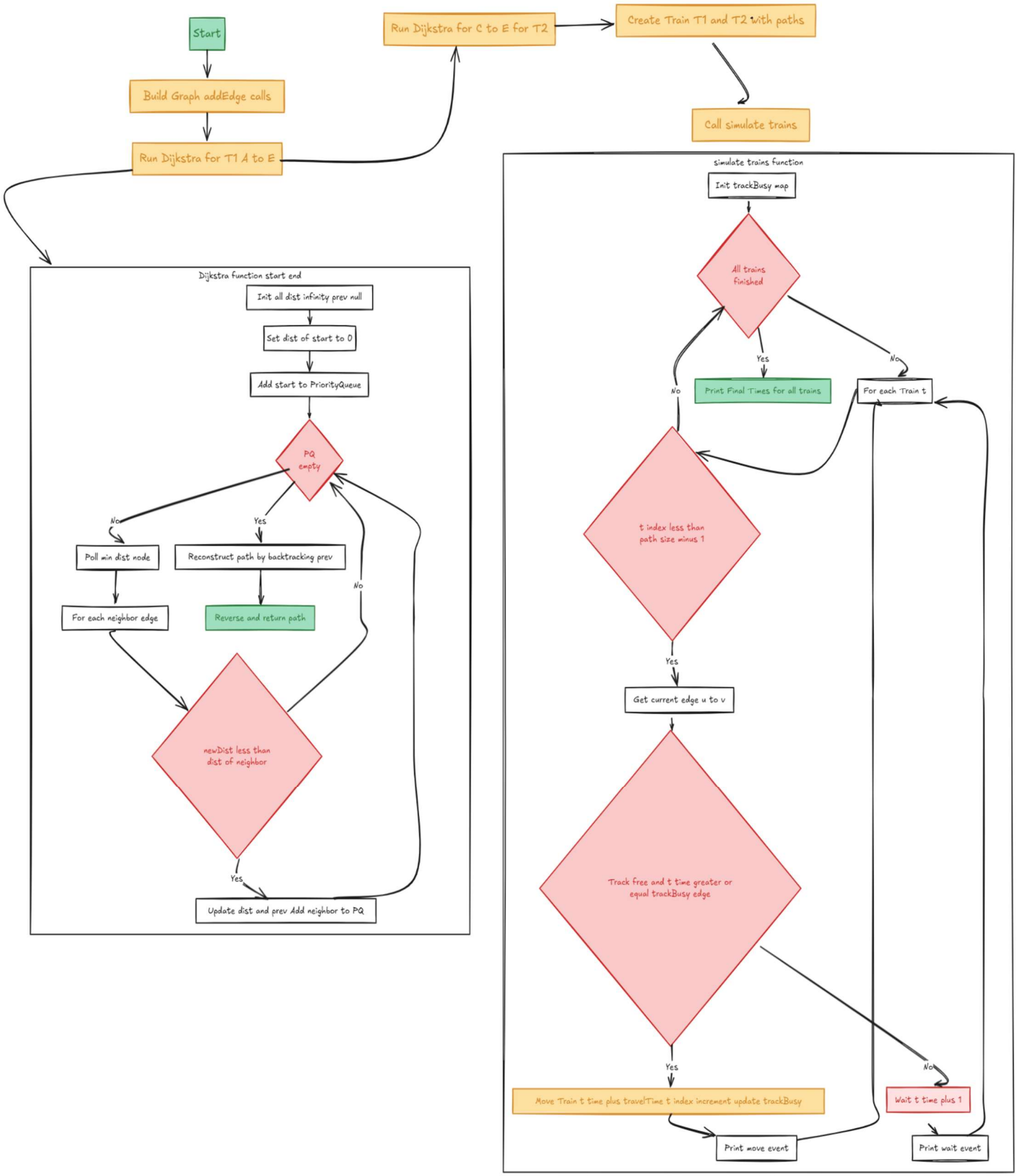
Used to resolve conflicts when multiple trains attempt to use the same track.

Rules:

Train with shortest remaining path gets priority

If equal, use first-come-first-serve

2.2 Flowchart



2.3 Pseudocode

```
Initialize graph
For each train:
        find shortest path using Dijkstra

While trains not finished:
        For each train:
                if next track is free:
                        move train
                else:
                        apply scheduling rule
                        wait if needed
Output total time and movement
```

Complexity Analysis

3.1 Time Complexity (Best, Average, Worst Case)

- Dijkstra: $O((V + E) \log V)$
- Scheduling (simulation): $O(T \times E)$
  Where:
  - V = number of stations
  - E = number of tracks
  - T = number of trains

3.2 Space Complexity

- Graph storage: $O(V + E)$
- Distance array: $O(V)$
- Train data: $O(T)$

Implementation

4.1 Programming Language Used

The project is implemented using Java, as it provides efficient data structures such as priority queue and hash maps.

4.2 Code Explanation

- Graph is stored using adjacency list
- Dijkstra algorithm computes shortest path
- Each train stores:
  - current position
  - path
  - time
- Conflict is handled using track occupancy:
- If track is free → train moves
- If occupied → train waits

Results and Discussion

5.1 Output

- Shortest path is computed for each train
- Trains move step-by-step
- Conflicts are detected and resolved
- Waiting time is recorded

5.2 Performance Analysis

- Efficient routing reduces travel time
- Greedy scheduling minimizes conflict
- System ensures better utilization of tracks

Limitations:

- Simple scheduling (not fully optimal)
- Small-scale simulation

Conclusion

This project successfully demonstrates how graph algorithms and scheduling techniques can be used to optimize train movement. By combining Dijkstra's algorithm with greedy scheduling, we reduce delays and improve efficiency. The approach is simple, effective, and suitable for small-scale systems.

Future Scope

- Implement dynamic scheduling using real-time data
- Use machine learning for smarter decisions
- Expand to large-scale railway networks
- Include bidirectional tracks and signals

References

1. Introduction to Algorithms – Cormen
2. GeeksforGeeks – Dijkstra Algorithm
3. Wikipedia – Graph Theory
4. Java Documentation – Priority Queue

GITHUB LINK - https://github.com/k4rtikx/Design-and-Analysis-of-Algorithms-Miniproject.git

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